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Options, Futures, and Other Derivatives
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Calculation of Moments for Valuing Basket Options

Consider the problem of calculating the first two moments of the value of a basket of assets at a future time, T , in a risk-neutral world. The price of each asset in the basket is assumed to be lognormal. Define

- n : The number of assets
- S_i : The value of the i th asset at time T ¹
- F_i : The forward price of the i th asset for a contract maturing at time T .
- σ_i : The volatility of the i th asset between time zero and time T
- ρ_{ij} : Correlation between returns from the i th and j th asset
- P : Value of basket at time T
- M_1 : First moment of P in a risk-neutral world
- M_2 : Second moment of P in a risk-neutral world

Because $P = \sum_{i=1}^n S_i$, $\hat{E}(S_i) = F_i$, $M_1 = \hat{E}(P)$ and $M_2 = \hat{E}(P^2)$ where $\hat{E}$ denotes expected value in a risk-neutral world, it follows that

$$M_1 = \sum_{i=1}^n F_i$$

Also,

$$P^2 = \sum_{i=1}^n \sum_{j=1}^n S_i S_j$$

From the properties of lognormal distributions

$$\hat{E}(S_i S_j) = F_i F_j e^{\rho_{ij} \sigma_i \sigma_j T}$$

Hence

$$M_2 = \sum_{i=1}^n \sum_{j=1}^n F_i F_j e^{\rho_{ij} \sigma_i \sigma_j T}$$

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¹ If the i th asset is a certain stock and there are, say, 200 shares of the stock in the basket, then the i th “asset” is defined as 200 shares of the stock and S_i is the value of 200 shares of the stock.